**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 043 · Copyright and Software Licences

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does the Computer Misuse Act 1990 address?

Q2. How is each bitmap pixel's colour stored?
A. As a sound sample
B. As a continuous voltage
C. As a binary code
D. As a network address

Q3. “RAM keeps every saved file permanently after shutdown.” Correct this. Explain why.

Q4. A mono recorder stores sound samples. Two recordings have the same sample rate and length but different bit depths. Correct both claims. Explain why. (a) Sample rate is bits per sample. (b) Bit depth is samples per second.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Copyright — Legal protection for original work including software.

Copyright, Designs and Patents Act 1988 — UK law protecting intellectual creations including computer programs.

Software licence — Conditions setting permitted ways to use, change or distribute software.

Open-source software — Software whose source code is available under an open-source licence.

Proprietary software — Software generally controlled by a rights holder with restricted source access.

Source code — Human-readable code written by programmers.

## New learning

Copyright protects creations: The Copyright, Designs and Patents Act 1988 protects many original works, including computer software. Copying or distributing protected software without appropriate permission or licence can infringe copyright.

Why licences matter: A software licence sets out permissions and conditions for using, copying, modifying and distributing software. Owning or downloading a copy does not mean you can do anything with it.

Open-source licensing: Open-source licences allow people to inspect source code and usually to modify and redistribute it under specified licence conditions. Open source does not necessarily mean the software has no cost or conditions.

Proprietary licensing: Proprietary software usually keeps the source code private and gives customers specified usage rights. It may offer vendor support, but users generally cannot modify or distribute the code without permission.

## Worked example

A club needs software it can adapt and inspect, but volunteers must follow licence rules. Explain why an open-source licence may suit it. [4 marks]

1. The club wants source-code access and customisation.

2. Open-source licensing provides access to source code [1] that can be inspected and modified [1].

3. This allows volunteers to adapt the software to their needs [1], provided they follow the terms of the chosen licence [1].

4. Avoid claiming all open-source software is free or that there are no conditions.

## We Do · Practise with guidance

### Guided 1

A museum wants to adapt the code of an exhibition app. Explain two benefits of open-source licensing. [4 marks]

Worked method: Access to source code [1] lets developers inspect how the app works [1]. Permission to modify under the licence [1] allows it to be adapted for the museum's exhibits [1].

### Guided 2

A company buys a proprietary graphics package with customer support. Explain one advantage and one disadvantage of this choice. [4 marks]

Worked method: Commercial/vendor support may be available [1], helping the company resolve issues [1]. Source code is usually unavailable [1], so the company cannot freely alter how the software works [1].

### Guided 3

A pupil copies a paid game and shares it with a class without permission. Explain why this may breach copyright and the role of a licence. [4 marks]

Worked method: The game is copyright-protected software [1]; copying/distributing it without permission may infringe the law [1]. The software licence specifies permitted use/distribution [1], and sharing outside those terms is not allowed [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Predict the software licence message.

```
mayModify = false
if mayModify then
    print("Edit permitted")
else
    print("Check licence")
endif
```

predict

### Question 1

State what a software licence does. [2 marks] Source: Original OCR J277-style teaching question; not an official OCR past paper.

### Question 2

A charity wants to change software to meet its accessibility needs. Recommend a licence type and explain your choice. [4 marks] Source: Original OCR J277-style teaching question; not an official OCR past paper.

### Question 3

Compare open-source and proprietary software for a small business, considering cost, modification and support. [6 marks] Source: Original OCR J277-style teaching question; not an official OCR past paper.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## Exit Ticket · Answer before checking

Open-source software may have legal licence conditions and is not automatically without cost.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. Unauthorised access to computer systems and related prohibited activities.

Q2. C — As a binary code

Q3. The program's active instructions and working data.

Q4. (a) Sample rate means samples per second: 8,000 Hz means 8,000 samples each second. A higher rate can add detail and use more space. (b) Bit depth is bits per sample; more bits allow more amplitude levels and use more space. Longer duration adds samples.

## I Do / We Do pages · Guided practice

Q1. Access to source code [1] lets developers inspect how the app works [1]. Permission to modify under the licence [1] allows it to be adapted for the museum's exhibits [1].

Q2. Commercial/vendor support may be available [1], helping the company resolve issues [1]. Source code is usually unavailable [1], so the company cannot freely alter how the software works [1].

Q3. The game is copyright-protected software [1]; copying/distributing it without permission may infringe the law [1]. The software licence specifies permitted use/distribution [1], and sharing outside those terms is not allowed [1].

## You Do pages · Code First

```
mayModify = false
if mayModify then
    print("Edit permitted")
else
    print("Check licence")
endif
```

Expected result: Check licence

## You Do pages · Independent questions

Q1. Sets out conditions or permissions for using software [1], including copying/modification/distribution rights [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q2. Open-source software can give source-code access [1] and allow changes under its licence [1]. The charity can customise the interface [1] to improve accessibility for its users [1]. Other justified options accepted if scenario supported.

Q3. Open-source licences provide source code [1], often allowing lawful modification [1]; licence terms still apply [1]. Proprietary software typically hides source [1], may charge licence fees [1], and may offer vendor support [1]. Justified alternatives about costs/support accepted.

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**